



Herramientas para la configuración y gestión de redes en sistemas Red Hat





Xavier Campos

Red Hat Academy Tech Lead y docente certificado

| PUE Academy





ESCENARIO DE TRABAJO

HERRAMIENTAS NECESARIAS

USO DE NMCLI

FICHEROS DEL SISTEMA

NMTUI

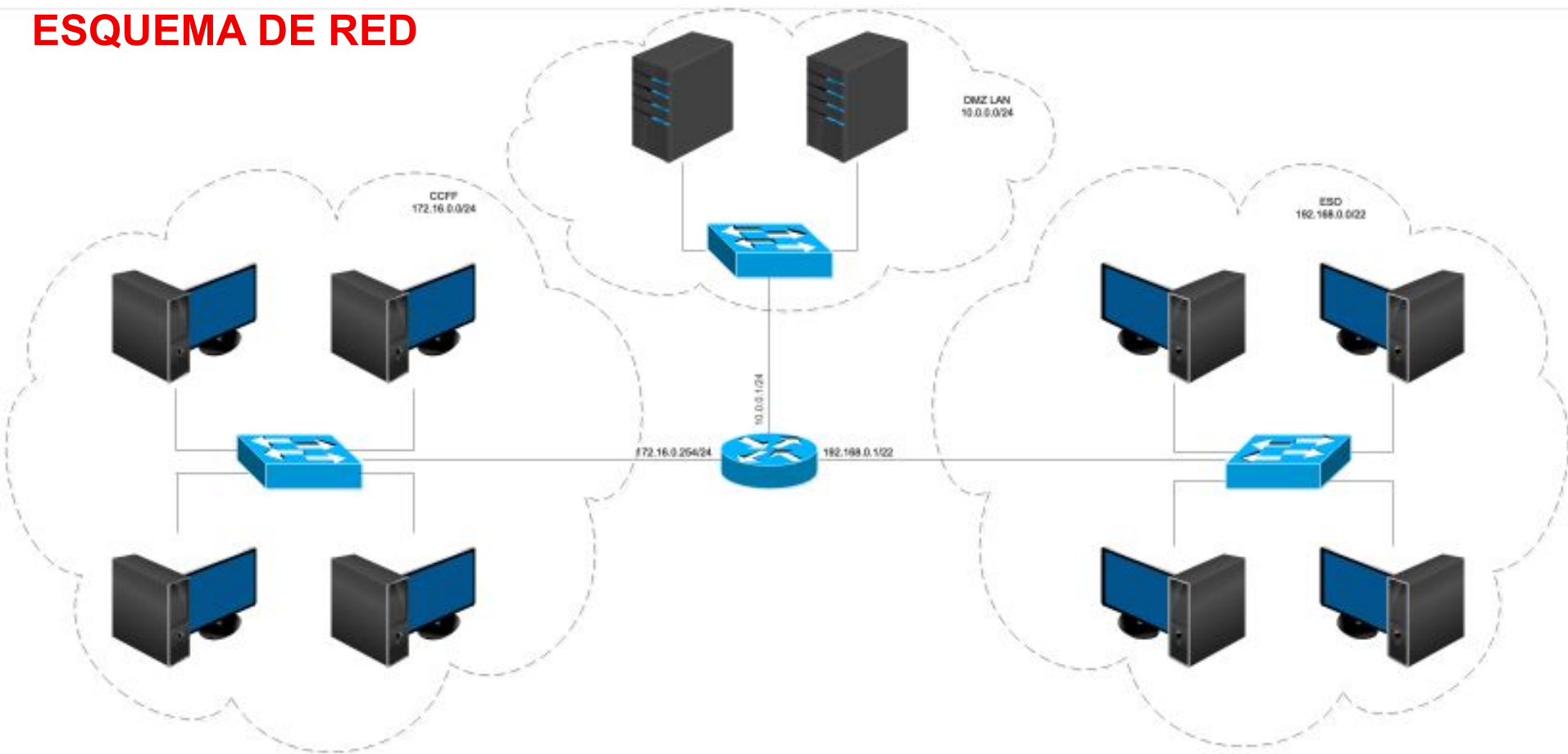
ESCENARIO DE TRABAJO

Vamos a utilizar como ejemplo un centro de educación secundaria en el que se encuentra la siguiente infraestructura de red:

- Red LAN para las aulas de ESO
- Red LAN para las aulas de CCFF
- Red LAN para los servidores propios del centro (DMZ)
- Red WLAN para los dispositivos inalámbricos (profesorado y despachos)



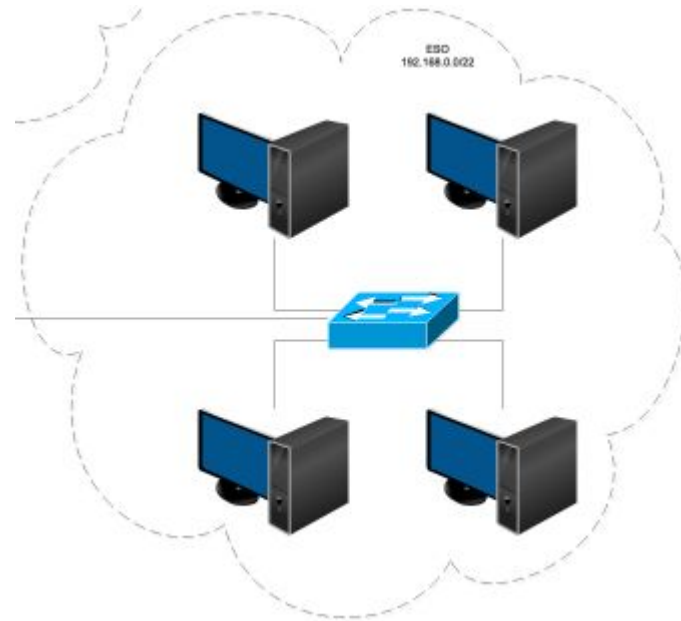
ESQUEMA DE RED



Red LAN para las aulas de ESO

Esta red tiene asignado el direccionamiento 192.168.0.0/22

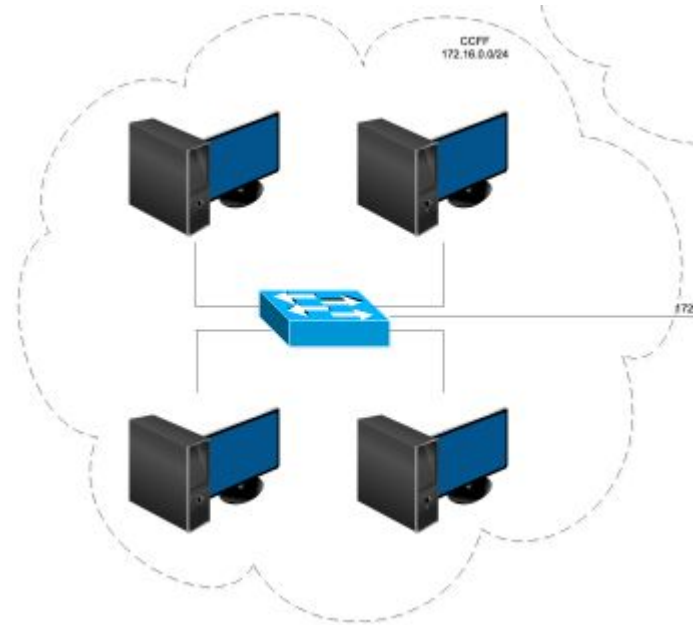
- Dirección de red: 192.168.0.0
- Máscara: 255.255.252.0
- Dirección de broadcast: 192.168.3.255
- Rango de direcciones: 192.168.0.1 - 192.168.3.254
- Núm. máximo de equipos: 1022



Red LAN para las aulas de CCFF

Esta red tiene asignado el direccionamiento 172.16.0.0/24

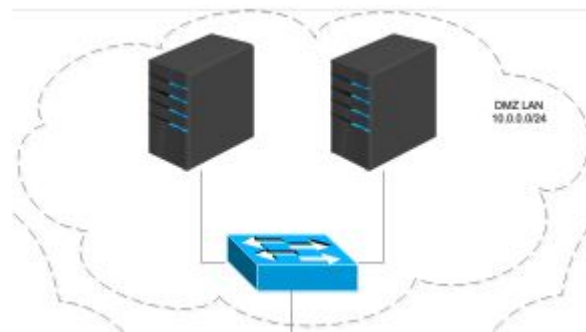
- Dirección de red: 172.16.0.0
- Máscara: 255.255.255.0
- Dirección de broadcast: 172.16.0.255
- Rango de direcciones: 172.16.0.1 - 172.16.0.254
- Núm. máximo de equipos: 254



Red LAN para la red de servidores

Esta red tiene asignado el direccionamiento 10.0.0.0/24

- Dirección de red: 10.0.0.0
- Máscara: 255.255.255.0
- Dirección de broadcast: 10.0.0.255
- Rango de direcciones: 10.0.0.1 - 10.0.0.254
- Núm. máximo de equipos: 254





ESCENARIO DE TRABAJO

HERRAMIENTAS NECESARIAS

USO DE NMCLI

FICHEROS DEL SISTEMA

NMTUI

HERRAMIENTAS NO NECESARIAS

➤ ifconfig

➤ netstat

➤ route

➤ arp

➤ traceroute

➤ nano

```
[root@rhel ~]# rpm -qf /usr/sbin/ifconfig
net-tools-2.0-0.51.20160912git.el8.x86_64
[root@rhel ~]#
[root@rhel ~]# rpm -qf /usr/bin/netstat
net-tools-2.0-0.51.20160912git.el8.x86_64
[root@rhel ~]#
[root@rhel ~]# rpm -qf /usr/sbin/route
net-tools-2.0-0.51.20160912git.el8.x86_64
[root@rhel ~]#
[root@rhel ~]# rpm -qf /usr/sbin/arp
net-tools-2.0-0.51.20160912git.el8.x86_64
[root@rhel ~]#
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[root@rhel ~]# ip link show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DE
    link/ether 00:1c:42:9e:5c:ee brd ff:ff:ff:ff:ff:ff
[root@rhel ~]# ip link set enp0s6 down
[root@rhel ~]# ip link show enp0s6
3: enp0s6: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT grou
    link/ether 00:1c:42:9e:5c:ee brd ff:ff:ff:ff:ff:ff
[root@rhel ~]# ip link set enp0s6 up
[root@rhel ~]# ip link show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DE
    link/ether 00:1c:42:9e:5c:ee brd ff:ff:ff:ff:ff:ff
[root@rhel ~]# ip -s link show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DE
    link/ether 00:1c:42:9e:5c:ee brd ff:ff:ff:ff:ff:ff
    RX: bytes  packets  errors  dropped  overrun  mcast
    2196      12        0        0        0        0
    TX: bytes  packets  errors  dropped  carrier  collsns
    0          0        0        0        0        0
[root@rhel ~]#
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[root@rhel ~]# ip neigh show
192.168.134.1 dev enp0s7 lladdr 00:1c:42:00:00:1d STALE
172.20.10.1 dev enp0s5 lladdr 84:aa:9c:4a:ba:94 REACHABLE
172.20.10.101 dev enp0s5 lladdr 78:7b:8a:dd:3e:18 REACHABLE
fe80::86aa:9cff:fe4a:ba94 dev enp0s5 lladdr 84:aa:9c:4a:ba:94 router STALE
fe80::21c:42ff:fe00:1d dev enp0s7 lladdr 00:1c:42:00:00:1d router STALE
[root@rhel ~]# ip neigh add 172.20.10.3 dev enp0s5 lladdr 00:0e:c6:79:73:e5
[root@rhel ~]# ip neigh show
192.168.134.1 dev enp0s7 lladdr 00:1c:42:00:00:1d STALE
172.20.10.3 dev enp0s5 lladdr 00:0e:c6:79:73:e5 PERMANENT
172.20.10.1 dev enp0s5 lladdr 84:aa:9c:4a:ba:94 REACHABLE
172.20.10.101 dev enp0s5 lladdr 78:7b:8a:dd:3e:18 REACHABLE
fe80::86aa:9cff:fe4a:ba94 dev enp0s5 lladdr 84:aa:9c:4a:ba:94 router STALE
fe80::21c:42ff:fe00:1d dev enp0s7 lladdr 00:1c:42:00:00:1d router STALE
[root@rhel ~]#
```



HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[root@rhel ~]# ip address show dev enp0s7
4: enp0s7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
    link/ether 00:1c:42:33:50:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.134.201/24 brd 192.168.134.255 scope global dynamic noprefixroute
        valid_lft 1645sec preferred_lft 1645sec
    inet6 fe80::f694:8852:728b:48f3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@rhel ~]# ip address add 10.0.0.99/24 dev enp0s7
[root@rhel ~]# ip address show dev enp0s7
4: enp0s7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
    link/ether 00:1c:42:33:50:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.134.201/24 brd 192.168.134.255 scope global dynamic noprefixroute
        valid_lft 1639sec preferred_lft 1639sec
    inet 10.0.0.99/24 scope global enp0s7
        valid_lft forever preferred_lft forever
    inet6 fe80::f694:8852:728b:48f3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@rhel ~]# █
```

HERRAMIENTAS NECESARIAS

▀ **ip**

▀ **ss**

▀ **ssh**

▀ **ping**

▀ **tracert / mtr**

▀ **vi**

```
[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
[root@rhel ~]#
[root@rhel ~]# ip route add 172.18.0.0/16 via 10.200.0.9
[root@rhel ~]#
[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
172.18.0.0/16 via 10.200.0.9 dev enp0s5
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
[root@rhel ~]#
[root@rhel ~]# ip route del 172.18.0.0/16
[root@rhel ~]#
[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
[root@rhel ~]# █
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[root@rhel ~]# ss -ltp
State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  users:(("dnsmasq",pid=1741,fd=6))
LISTEN  0        32      192.168.122.1:domain  0.0.0.0:*          users:(("sshd",pid=817,fd=6))
LISTEN  0        128      0.0.0.0:ssh          0.0.0.0:*          users:(("cupsd",pid=820,fd=10))
LISTEN  0         5       127.0.0.1:ipp         0.0.0.0:*          users:(("rpcbind",pid=654,fd=4),("systemd",pid=1,fd=39))
LISTEN  0       128      0.0.0.0:sunrpc        0.0.0.0:*          users:(("sshd",pid=817,fd=8))
LISTEN  0       128      [::]:ssh              [::]:*             users:(("cupsd",pid=820,fd=9))
LISTEN  0         5       [::1]:ipp              [::]:*             users:(("rpcbind",pid=654,fd=6),("systemd",pid=1,fd=41))
LISTEN  0       128      [::]:sunrpc           [::]:*
[root@rhel ~]#
```


HERRAMIENTAS NECESARIAS

▀ ip

▀ **ss**

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

Opción	Descripción
-n	Muestra números en lugar de nombres para las interfaces y los puertos.
-t	Muestra los sockets TCP.
-u	Muestra los sockets UDP.
-l	Muestra solo los sockets a los que está atento.
-a	Muestra todos los sockets (los que escucha y los establecidos).
-p	Muestra el proceso de usar los sockets.

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[root@rhel ~]# ssh super@localhost
super@localhost's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Oct  4 14:13:34 2021 from ::1
[super@rhel ~]$ exit
logout
Connection to localhost closed.
[root@rhel ~]# ssh localhost
root@localhost's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Oct  4 06:55:02 2021
[root@rhel ~]# exit
logout
Connection to localhost closed.
[root@rhel ~]# █
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[root@rhel ~]# ssh red_hat
super@localhost's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Oct  4 14:16:47 2021 from ::1
[super@rhel ~]$ exit
logout
Connection to localhost closed.
[root@rhel ~]# cat .ssh/config
Host red_hat
    User super
    Hostname localhost
    Port 22
[root@rhel ~]# █
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[root@rhel ~]# cat .ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQAC2uySlfMmw/nqJqR88ZcICSXBY7IJFr9WEIUwt/r4rZDb7
HzdWnlwtcMtrbXt7DgT7ZValtKMvTrvA1mY+Siwd50HFfM4AyXM8T7HZEiPT//CAo8PKTCUnhqDQD1kI+L
3e7zwnPECMBH1+wXVAUwHk8JJJoZEXet1gTBerxujS785hmXqLrIgJvYefjp7K1cCEZ8tBPYELMAPZX0E6H+c
[root@rhel ~]#
[root@rhel ~]# ssh red_hat
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Oct  4 14:29:03 2021 from ::1
[super@rhel ~]$
[super@rhel ~]$ cat .ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQAC2uySlfMmw/nqJqR88ZcICSXBY7IJFr9WEIUwt/r4rZDb7
HzdWnlwtcMtrbXt7DgT7ZValtKMvTrvA1mY+Siwd50HFfM4AyXM8T7HZEiPT//CAo8PKTCUnhqDQD1kI+L
3e7zwnPECMBH1+wXVAUwHk8JJJoZEXet1gTBerxujS785hmXqLrIgJvYefjp7K1cCEZ8tBPYELMAPZX0E6H+c
[super@rhel ~]$
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
[super@rhel bin]$ ping -c4 redhat.com
PING redhat.com (209.132.183.105) 56(84) bytes of data.
64 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=1 ttl=233 time=166 ms
64 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=2 ttl=233 time=166 ms
64 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=3 ttl=233 time=166 ms
64 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=4 ttl=233 time=166 ms

--- redhat.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 10ms
rtt min/avg/max/mdev = 165.802/166.212/166.477/0.260 ms
[super@rhel bin]$
[super@rhel bin]$ ping -q -c4 redhat.com
PING redhat.com (209.132.183.105) 56(84) bytes of data.

--- redhat.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 9ms
rtt min/avg/max/mdev = 165.676/165.824/165.909/0.301 ms
[super@rhel bin]$ █
```

HERRAMIENTAS NECESARIAS

➤ ip

➤ ss

➤ ssh

➤ ping

➤ tracepath / mtr

➤ vi

```
[root@rhel ~]# ping -c4 -i 0.1 -s 1500 redhat.com
PING redhat.com (209.132.183.105) 1500(1528) bytes of data.
1508 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=1 ttl=127 time=170 n
1508 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=2 ttl=127 time=228 n
1508 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=3 ttl=127 time=170 n
1508 bytes from redirect.redhat.com (209.132.183.105): icmp_seq=4 ttl=127 time=170 n

--- redhat.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 316ms
rtt min/avg/max/mdev = 169.707/184.434/227.730/24.998 ms, pipe 3
[root@rhel ~]#
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ SS

▀ ssh

▀ ping

▀ tracpath / mtr

▀ vi

```
[root@rhel ~]# tracpath -n 172.20.10.1
1?: [LOCALHOST] pmtu 1500
1: 192.168.124.254 0.664ms
1: 192.168.124.254 0.605ms
2: 172.20.10.1 1.979ms reached
Resume: pmtu 1500 hops 2 back 2
[root@rhel ~]#
```

```
My traceroute [v0.87]
rhel (0.0.0.0) Tue Oct 5 05:03:37 2021
Keys: Help Display mode Restart statistics Order of fields quit
```

Host	Packets		Pings				
	Loss%	Snt	Last	Avg	Best	Wrst	StDev
1. 158.69.192.1	0.0%	33	0.2	0.1	0.1	0.2	0.0
2. 192.168.143.254	0.0%	33	0.2	0.2	0.1	0.3	0.0
3. 149.56.50.190	0.0%	33	0.3	0.3	0.2	0.4	0.0
4. 10.98.243.65	0.0%	33	0.2	0.3	0.2	0.4	0.0
5. be121.bhs-d1-a75.qc.ca	0.0%	32	0.3	0.3	0.2	0.5	0.0
6. 10.95.82.1	0.0%	32	0.4	0.3	0.2	0.5	0.0
7. ???							
8. 10.73.48.30	0.0%	32	84.4	84.4	84.3	84.5	0.0
9. be7.p19-2-6k.fr.eu	0.0%	32	81.4	89.3	81.3	189.4	23.7
10. be32.p19-55-6k.fr.eu	0.0%	32	85.1	84.9	84.5	85.6	0.0
11. ouaib.ovh.net	0.0%	32	81.6	82.4	81.5	97.4	3.1

HERRAMIENTAS NECESARIAS

- ▶ ip
- ▶ ss
- ▶ ssh
- ▶ ping
- ▶ tracepath / mtr
- ▶ vi

```
=== Desplazamiento ===  
- h,j,k,l : izda,abajo,arriba,derecha  
- w : salto hacia delante entre "palabras"  
- W : salto hacia delante entre "espacios"  
- b : salto hacia atrás entre "palabras"  
- B : salto hacia atrás entre "espacios"  
- [N linea]G : ir a la línea N  
- G : ir a la última línea  
- gg : ir a la primera línea  
- r : reemplazar el carácter sobre el cursor sin entrar en navegación  
- u : deshacer última operación  
- x : suprimir el carácter bajo el cursor (excepto salto de línea)  
- X : suprimir el carácter a la izda el cursor
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
==== Modo inserción ====
- i : Entrar en el modo de inserción en el lugar del cursor.
- a : Entrar en el modo de inserción después del cursor.
- o : Entrar en el modo de inserción en la línea siguiente.
- I : Entrar en el modo de inserción al inicio de línea.
- A : Entrar en el modo de inserción al final de línea.
- O : Entrar en el modo de inserción en la línea anterior.
```


HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
=== Búsqueda ===
- / : Buscar hacia abajo
- ? : Buscar hacia arriba
- n : siguiente coincidencia
- N : previa coincidencia

=== Manipulación ===
- d[N]d : cortar líneas
- d[N]w : cortar palabras
- y[N]y : copiar líneas
- y[N]w : copiar palabras
- p : pegar debajo de la línea actual
- P : pegar encima de la línea actual
```

HERRAMIENTAS NECESARIAS

▀ ip

▀ ss

▀ ssh

▀ ping

▀ tracepath / mtr

▀ vi

```
=== Ejecución ===
```

```
- w : guardar cambios  
- q : salir  
- q! : salir sin guardar  
- wq : guardar y salir  
- w [RUTA] : guardar como ...  
- ! : ejecutar un comando  
- r! : ejecutar un comando y redirigir stdout al fichero que estamos editando  
- s/patternA/patternB/ : sustituciones de //patternA// por //patternB//
```



ESCENARIO DE TRABAJO

HERRAMIENTAS NECESARIAS

USO DE NMCLI

FICHEROS DEL SISTEMA

NMTUI

NMCLI - Dispositivos vs Conexiones

- Herramienta de consola que permite gestionar las configuraciones del servicio Network Manager
- Permite la gestión y administración de conexiones.
- Una conexión es un conjunto de parámetros definidos para su uso sobre un dispositivo de la máquina
- nmcli identifica como “dispositivo” a cualquier interfaz de red que el SO detecta en el equipo
- Un dispositivo puede tener asociadas varias conexiones, pero sólo una puede estar activa.

```
[root@rhel ~]# nmcli device status
```

DEVICE	TYPE	STATE	CONNECTION
enp0s5	ethernet	conectado	enp0s5
enp0s7	ethernet	conectado	Conexión cableada 1
virbr0	bridge	conectado	virbr0
enp0s6	ethernet	desconectado	--
lo	loopback	sin gestión	--
virbr0-nic	tun	sin gestión	--

```
[root@rhel ~]#
```

```
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
Conexión cableada 1	6eaf201c-ee84-3d25-bceb-4af4b76b0d26	ethernet	enp0s7
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
virbr0	537d3b4b-4f34-47fb-a38b-a47df1acded0	bridge	virbr0

```
[root@rhel ~]#
```

```
[root@rhel ~]#
```

```
[root@rhel ~]#
```

```
[root@rhel ~]#
```

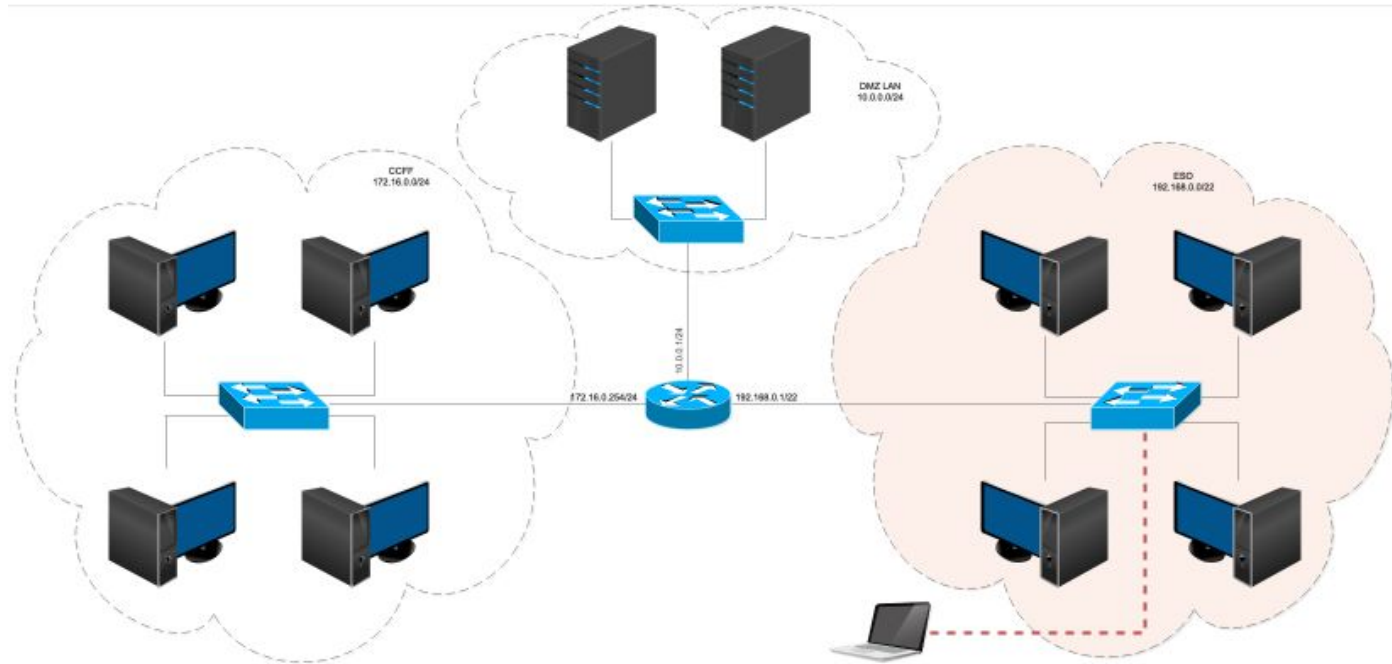
NMCLI - Nueva conexión

```
[root@rhel ~]# nmcli connection add
```

autoconnect	connection.stable-id	ipv4.ignore-auto-routes	ipv6.may-fail
<u>con-name</u>	connection.timestamp	ipv4.may-fail	ipv6.method
connection.auth-retries	connection.type	<u>ipv4.method</u>	ipv6.never-default
<u>connection.autoconnect</u>	connection.uuid	ipv4.never-default	ipv6.route-metric
connection.autoconnect-priority	connection.zone	ipv4.route-metric	ipv6.routes
connection.autoconnect-retries	help	ipv4.routes	ipv6.route-table
connection.autoconnect-slaves	<u>ifname</u>	ipv4.route-table	ipv6.token
connection.gateway-ping-timeout	<u>ipv4.addresses</u>	ipv6.addresses	master
connection.id	ipv4.dad-timeout	ipv6.addr-gen-mode	match.interface-name
connection.interface-name	ipv4.dhcp-client-id	ipv6.dhcp-duid	proxy.browser-only
connection.lldp	ipv4.dhcp-fqdn	ipv6.dhcp-hostname	proxy.method
connection.llmnr	ipv4.dhcp-hostname	ipv6.dhcp-send-hostname	proxy.pac-script
connection.master	ipv4.dhcp-send-hostname	ipv6.dns	proxy.pac-url
connection.mdns	ipv4.dhcp-timeout	ipv6.dns-options	slave-type
connection.metered	<u>ipv4.dns</u>	ipv6.dns-priority	tc.qdiscs
connection.multi-connect	ipv4.dns-options	ipv6.dns-search	tc.tfilters
connection.permissions	ipv4.dns-priority	ipv6.gateway	<u>type</u>
connection.read-only	ipv4.dns-search	ipv6.ignore-auto-dns	
connection.secondaries	<u>ipv4.gateway</u>	ipv6.ignore-auto-routes	
connection.slave-type	ipv4.ignore-auto-dns	ipv6.ip6-privacy	

NMCLI - Conexión a la red “ESO”

- Red: 192.168.0.0/22
- DHCP: No
- IP: 192.168.0.99
- GW: 192.168.0.1
- DNS: 213.176.161.16



NMCLI - Conexión a la red “ESO”

```
[root@rhel ~]# nmcli connection add con-name "ESO" ifname
enp0s5      enp0s6      lo      virbr0      virbr0-nic
[root@rhel ~]# nmcli connection add con-name "ESO" ifname enp0s6 type
6lowpan      bluetooth      ethernet      macsec      ovs-port      vpn
802-11-olpc-mesh bond      generic      macvlan      pppoe      vxlan
802-11-wireless bridge      gsm      olpc-mesh      team      wifi
802-3-ethernet cdma      infiniband      ovs-bridge      tun      wimax
adsl      dummy      ip-tunnel      ovs-interface      vlan      wpan
[root@rhel ~]# nmcli connection add con-name "ESO" ifname enp0s6 type ethernet connection.autoconnect
no yes
[root@rhel ~]# nmcli connection add con-name "ESO" ifname enp0s6 type ethernet connection.autoconnect no \
> ipv4.method manual ipv4.addresses 192.168.0.99/22 ipv4.gateway 192.168.0.1 ipv4.dns 213.176.161.16
Conexión 'ESO' (317f33ef-62b7-4ceb-805d-729e124221dc) agregada con éxito.
[root@rhel ~]#
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
ESO	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	--

```
[root@rhel ~]# █
```


NMCLI - Conexión a la red “ESO”

```
[root@rhel ~]# nmcli device status
DEVICE      TYPE      STATE      CONNECTION
enp0s5      ethernet  conectado  enp0s5
virbr0      bridge   conectado  virbr0
enp0s6      ethernet  desconectado --
lo          loopback  sin gestión --
virbr0-nic  tun      sin gestión --
[root@rhel ~]#
[root@rhel ~]# nmcli connection show
NAME        UUID                                  TYPE      DEVICE
enp0s5      f1544463-ec63-4907-ab80-5a0ac651a429 ethernet  enp0s5
virbr0      a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802 bridge    virbr0
ESO         317f33ef-62b7-4ceb-805d-729e124221dc ethernet  --
[root@rhel ~]#
[root@rhel ~]# ip address show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:1c:42:a3:23:39 brd ff:ff:ff:ff:ff:ff
[root@rhel ~]#
[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
[root@rhel ~]#
```

NMCLI - Conexión a la red “ESO”

```
[root@rhel ~]# nmcli connection up ESO
Conexión activada con éxito (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
[root@rhel ~]# nmcli device status
```

DEVICE	TYPE	STATE	CONNECTION
enp0s5	ethernet	conectado	enp0s5
enp0s6	ethernet	conectado	ESO
virbr0	bridge	conectado	virbr0
lo	loopback	sin gestión	--
virbr0-nic	tun	sin gestión	--

```
[root@rhel ~]# nmcli connection show
```

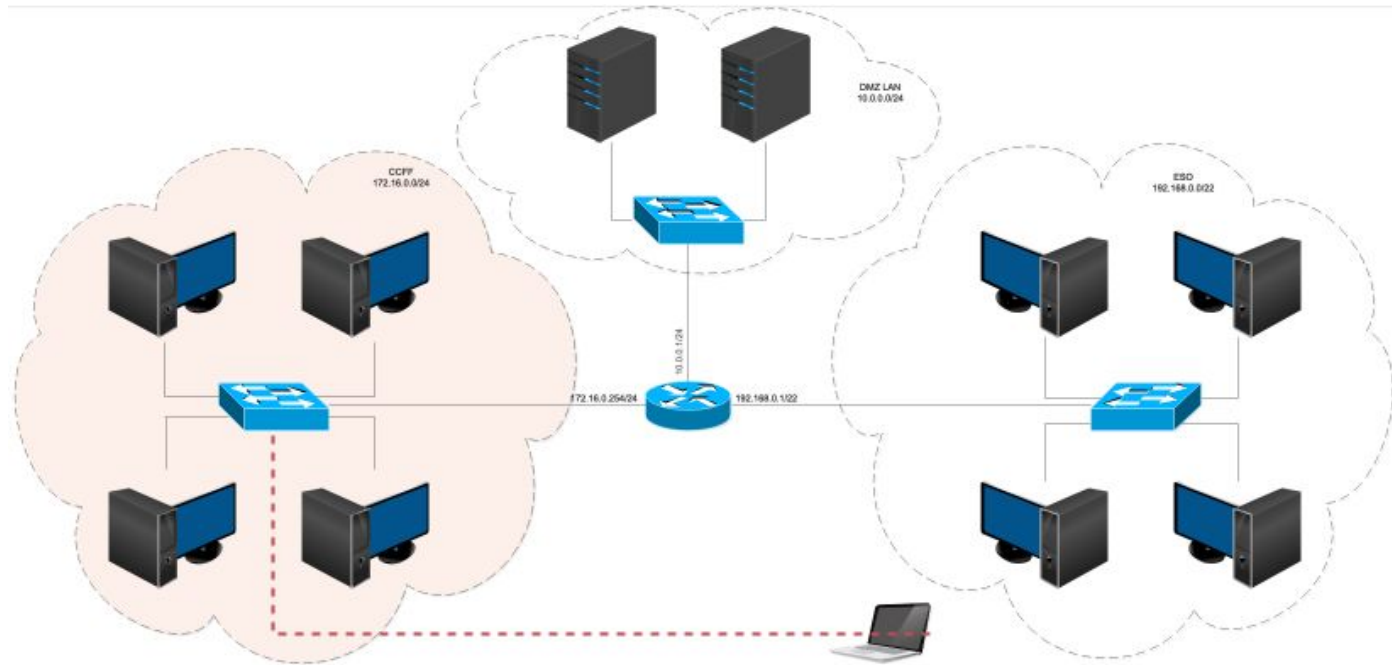
NAME	UUID	TYPE	DEVICE
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
ESO	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	enp0s6
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0

```
[root@rhel ~]# ip -4 address show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 192.168.0.99/22 brd 192.168.3.255 scope global noprefixroute enp0s6
        valid_lft forever preferred_lft forever
[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
default via 192.168.0.1 dev enp0s6 proto static metric 101
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
192.168.0.0/22 dev enp0s6 proto kernel scope link src 192.168.0.99 metric 101
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
```

NMCLI - Conexión a la red “CCFF”

► Red: 172.16.0.0/24

► DHCP: Sí



NMCLI - Conexión a la red "CCFF"

```
[root@rhel ~]# nmcli connection add con-name "CCFF" ifname
enp0s5      enp0s6      lo      virbr0      virbr0-nic
[root@rhel ~]# nmcli connection add con-name "CCFF" ifname enp0s6 type
6lowpan      bluetooth      ethernet      macsec      ovs-port      vpn
802-11-olpc-mesh bond      generic      macvlan      pppoe      vxlan
802-11-wireless bridge      gsm      olpc-mesh      team      wifi
802-3-ethernet cdma      infiniband      ovs-bridge      tun      wimax
adsl      dummy      ip-tunnel      ovs-interface      vlan      wpan
[root@rhel ~]# nmcli connection add con-name "CCFF" ifname enp0s6 type ethernet connection.autoconnect
no yes
[root@rhel ~]# nmcli connection add con-name "CCFF" ifname enp0s6 type ethernet connection.autoconnect yes ipv4.method
auto disabled link-local manual shared
[root@rhel ~]# nmcli connection add con-name "CCFF" ifname enp0s6 type ethernet connection.autoconnect yes ipv4.method auto
Conexión 'CCFF' (d7dfe82a-4f63-446d-9b4e-6c5c203b8432) agregada con éxito.
[root@rhel ~]#
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
ES0	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	enp0s6
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
CCFF	d7dfe82a-4f63-446d-9b4e-6c5c203b8432	ethernet	--

```
[root@rhel ~]# █
```

NMCLI - Conexión a la red “CCFF”

```
[root@rhel ~]# nmcli device status
DEVICE      TYPE      STATE      CONNECTION
enp0s5      ethernet  conectado  enp0s5
enp0s6      ethernet  conectado  ESO
virbr0      bridge   conectado  virbr0
lo          loopback  sin gestión --
virbr0-nic  tun      sin gestión --

[root@rhel ~]# nmcli connection show
NAME        UUID                                TYPE      DEVICE
enp0s5      f1544463-ec63-4907-ab80-5a0ac651a429  ethernet  enp0s5
ESO         317f33ef-62b7-4ceb-805d-729e124221dc  ethernet  enp0s6
virbr0      a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802  bridge    virbr0
CCFF        d7dfe82a-4f63-446d-9b4e-6c5c203b8432  ethernet  --

[root@rhel ~]# ip -4 address show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 192.168.0.99/22 brd 192.168.3.255 scope global noprefixroute enp0s6
        valid_lft forever preferred_lft forever

[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
default via 192.168.0.1 dev enp0s6 proto static metric 101
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
192.168.0.0/22 dev enp0s6 proto kernel scope link src 192.168.0.99 metric 101
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown

[root@rhel ~]#
```


NMCLI - Conexión a la red “CCFF”

```
[root@rhel ~]# nmcli connection up CCFF
Conexión activada con éxito (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/10)
[root@rhel ~]# nmcli device status
```

DEVICE	TYPE	STATE	CONNECTION
enp0s5	ethernet	conectado	enp0s5
enp0s6	ethernet	conectado	CCFF
virbr0	bridge	conectado	virbr0
lo	loopback	sin gestión	--
virbr0-nic	tun	sin gestión	--

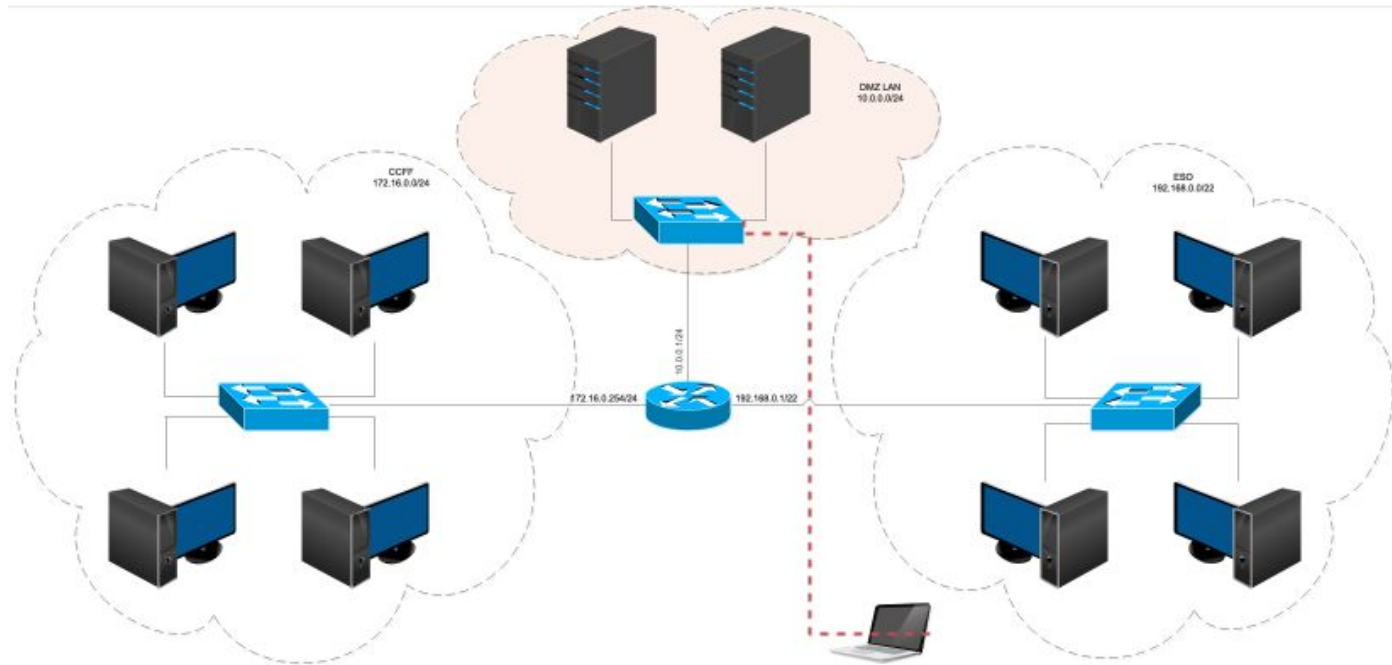
```
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
CCFF	d7dfe82a-4f63-446d-9b4e-6c5c203b8432	ethernet	enp0s6
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
ES0	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	--

```
[root@rhel ~]# ip -4 address show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 172.16.0.151/24 brd 172.16.0.255 scope global dynamic noprefixroute enp0s6
        valid_lft 35991sec preferred_lft 35991sec
[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
default via 172.16.0.254 dev enp0s6 proto dhcp metric 101
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
172.16.0.0/24 dev enp0s6 proto kernel scope link src 172.16.0.151 metric 101
```

NMCLI - Conexión a la red “DMZ”

- Red: 10.0.0.0/24
- DHCP: No
- IP: 10.0.0.99
- GW: 10.0.0.1



NMCLI - Conexión a la red “DMZ”

```
[root@rhel ~]# nmcli connection add con-name "DMZ" ifname
enp0s5      enp0s6      lo      virbr0      virbr0-nic
[root@rhel ~]# nmcli connection add con-name "DMZ" ifname enp0s6 type
6lowpan      bluetooth      ethernet      macsec      ovs-port      vpn
802-11-olpc-mesh bond      generic      macvlan      pppoe      vxlan
802-11-wireless bridge      gsm      olpc-mesh      team      wifi
802-3-ethernet cdma      infiniband      ovs-bridge      tun      wimax
adsl      dummy      ip-tunnel      ovs-interface      vlan      wpan
[root@rhel ~]# nmcli connection add con-name "DMZ" ifname enp0s6 type ethernet connection.autoconnect
no yes
[root@rhel ~]# nmcli connection add con-name "DMZ" ifname enp0s6 type ethernet connection.autoconnect no \
> ipv4.method manual ipv4.addresses 10.0.0.99/24
Conexión 'DMZ' (8769b2de-e33d-42a9-bf05-728da297561c) agregada con éxito.
[root@rhel ~]#
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
ES0	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	enp0s6
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
CCFF	d7dfe82a-4f63-446d-9b4e-6c5c203b8432	ethernet	--
DMZ	8769b2de-e33d-42a9-bf05-728da297561c	ethernet	--

```
[root@rhel ~]# █
```


NMCLI - Conexión a la red “DMZ”

```
[root@rhel ~]# nmcli device status
DEVICE      TYPE      STATE      CONNECTION
enp0s5      ethernet  conectado  enp0s5
enp0s6      ethernet  conectado  ESO
virbr0      bridge   conectado  virbr0
lo          loopback  sin gestión --
virbr0-nic  tun      sin gestión --

[root@rhel ~]# nmcli connection show
NAME        UUID                                  TYPE      DEVICE
enp0s5      f1544463-ec63-4907-ab80-5a0ac651a429  ethernet  enp0s5
ESO         317f33ef-62b7-4ceb-805d-729e124221dc  ethernet  enp0s6
virbr0      a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802  bridge    virbr0
CCFF        d7dfe82a-4f63-446d-9b4e-6c5c203b8432  ethernet  --
DMZ         8769b2de-e33d-42a9-bf05-728da297561c  ethernet  --

[root@rhel ~]# ip -4 address show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 192.168.0.99/22 brd 192.168.3.255 scope global noprefixroute enp0s6
        valid_lft forever preferred_lft forever

[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
default via 192.168.0.1 dev enp0s6 proto static metric 101
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
192.168.0.0/22 dev enp0s6 proto kernel scope link src 192.168.0.99 metric 101
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
```

NMCLI - Conexión a la red “DMZ”

```
[root@rhel ~]# nmcli connection up DMZ
Conexión activada con éxito (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/12)
[root@rhel ~]# nmcli device status
```

DEVICE	TYPE	STATE	CONNECTION
enp0s5	ethernet	conectado	enp0s5
enp0s6	ethernet	conectado	DMZ
virbr0	bridge	conectado	virbr0
lo	loopback	sin gestión	--
virbr0-nic	tun	sin gestión	--

```
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
DMZ	8769b2de-e33d-42a9-bf05-728da297561c	ethernet	enp0s6
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
CCFF	d7dfe82a-4f63-446d-9b4e-6c5c203b8432	ethernet	--
ES0	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	--

```
[root@rhel ~]# ip -4 address show enp0s6
3: enp0s6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 10.0.0.99/24 brd 10.0.0.255 scope global noprefixroute enp0s6
        valid_lft forever preferred_lft forever
[root@rhel ~]# ip route show
default via 10.200.0.1 dev enp0s5 proto dhcp metric 100
10.0.0.0/24 dev enp0s6 proto kernel scope link src 10.0.0.99 metric 101
10.200.0.0/24 dev enp0s5 proto kernel scope link src 10.200.0.205 metric 100
```

NMCLI - Modificación de una conexión

```
[root@rhel ~]# nmcli connection show ESO
connection.id:          ESO
connection.uuid:        317f33ef-62b7-4ceb-805d-729e124221dc
connection.stable-id:   --
connection.type:        802-3-ethernet
connection.interface-name: enp0s6
connection.autoconnect: no
...
...
...
ipv4.method:            manual
ipv4.dns:                213.176.161.16
ipv4.dns-search:        --
ipv4.dns-options:       ""
ipv4.dns-priority:      0
ipv4.addresses:          192.168.0.99/22
ipv4.gateway:            192.168.0.1
...
...
...
[root@rhel ~]#
```

NMCLI - Modificación de una conexión

```
[root@rhel ~]# nmcli connection modify id
CCFF DMZ enp0s5 ESO virbr0
[root@rhel ~]# nmcli connection modify id ESO ipv4.dns 213.176.161.18
[root@rhel ~]# nmcli connection show ESO
connection.id:          ESO
connection.uuid:        317f33ef-62b7-4ceb-805d-729e124221dc
connection.stable-id:   --
connection.type:        802-3-ethernet
connection.interface-name: enp0s6
connection.autoconnect: no
...
...
...
ipv4.method:            manual
ipv4.dns:                213.176.161.18
ipv4.dns-search:        --
ipv4.dns-options:       ""
ipv4.dns-priority:      0
ipv4.addresses:          192.168.0.99/22
ipv4.gateway:            192.168.0.1
...
...
...
[root@rhel ~]#
```

NMCLI - Modificación de una conexión

```
[root@rhel ~]# nmcli connection modify id ESO +ipv4.dns 213.176.161.16
[root@rhel ~]# nmcli connection show ESO
connection.id:                ESO
connection.uuid:              317f33ef-62b7-4ceb-805d-729e124221dc
connection.stable-id:        --
connection.type:              802-3-ethernet
connection.interface-name:    enp0s6
connection.autoconnect:      no
...
...
...
ipv4.method:                  manual
ipv4.dns:                     213.176.161.18,213.176.161.16
ipv4.dns-search:              --
ipv4.dns-options:             ""
ipv4.dns-priority:            0
ipv4.addresses:               192.168.0.99/22
ipv4.gateway:                 192.168.0.1
...
...
...

[root@rhel ~]#
```

NMCLI - Eliminación de una conexión

```
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
ES0	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	enp0s6
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
CCFF	d7dfe82a-4f63-446d-9b4e-6c5c203b8432	ethernet	--
DMZ	8769b2de-e33d-42a9-bf05-728da297561c	ethernet	--

```
[root@rhel ~]# nmcli connection delete DMZ
Conexión 'DMZ' (8769b2de-e33d-42a9-bf05-728da297561c) borrada correctamente.
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
ES0	317f33ef-62b7-4ceb-805d-729e124221dc	ethernet	enp0s6
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
CCFF	d7dfe82a-4f63-446d-9b4e-6c5c203b8432	ethernet	--

```
[root@rhel ~]#
```



ESCENARIO DE TRABAJO

HERRAMIENTAS NECESARIAS

USO DE NMCLI

FICHEROS DEL SISTEMA

NMTUI

/etc/sysconfig/network-scripts/

```
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
DMZ	704b9083-ee50-4e29-95c3-0bdd0b055292	ethernet	enp0s6
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
virbr0	a3913ff4-3fb0-44fd-9f6e-aa6a9aaa2802	bridge	virbr0
CCFF	d9f4e423-df5c-45ab-8773-321e3d51f384	ethernet	--
ES0	c4963e7f-e4e0-405f-8db6-68318a60d883	ethernet	--

```
[root@rhel ~]#  
[root@rhel ~]# tree /etc/sysconfig/network-scripts/  
/etc/sysconfig/network-scripts/  
├─ ifcfg-CCFF  
├─ ifcfg-DMZ  
├─ ifcfg-enp0s5  
└─ ifcfg-ES0  
  
0 directories, 4 files  
[root@rhel ~]#
```

/etc/sysconfig/network-scripts/

```
[root@rhel ~]# cat /etc/sysconfig/network-scripts/ifcfg-ES0
TYPE=Ethernet
PROXY_METHOD=none
BROWSER_ONLY=no
BOOTPROTO=dhcp
IPADDR=192.168.0.99
PREFIX=22
GATEWAY=192.168.0.1
DNS1=213.176.161.16
DEFROUTE=yes
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
IPV6_ADDR_GEN_MODE=stable-privacy
NAME=ES0
UUID=c4963e7f-e4e0-405f-8db6-68318a60d883
DEVICE=enp0s6
ONBOOT=no
[root@rhel ~]# █
```

Equivalencias de configuración

nmcli con mod	ifcfg-* file	Efecto
ipv4.method manual	BOOTPROTO=none	Las direcciones IPv4 se configuran de manera estática.
ipv4.method auto	BOOTPROTO=dhcp	Busca parámetros de configuración de un servidor DHCPv4. Si también hay direcciones estáticas configuradas, no las activará sin antes obtener información del servidor DHCPv4.
ipv4.addresses "192.0.2.1/24 192.0.2.254"	IPADDR0=192.0.2.1 PREFIX0=24 GATEWAY0=192.0.2.254	Establece direcciones IPv4 estáticas, prefijos de red y puertas de enlace predeterminadas. Si se configura más de un parámetro para la conexión, en lugar de terminar en 0, las directivas <code>ifcfg-*</code> terminarán en 1, 2, 3 y así sucesivamente.
ipv4.dns 8.8.8.8	DNS0=8.8.8.8	Modifica el archivo <code>/etc/resolv.conf</code> para usar este servidor de nombres.

Equivalencias de configuración

nmcli con mod	ifcfg-* file	Efecto
ipv4.dns-search example.com	DOMAIN=example.com	Modifica <code>/etc/resolv.conf</code> para que use este dominio en la directiva <code>buscar</code> .
connection.autoconnect yes	ONBOOT=yes	Activa automáticamente esta conexión durante el arranque.
connection.id ens3	NAME=ens3	El nombre de esta conexión.
connection.interface-name ens3	DEVICE=ens3	La conexión se limita a la interfaz de red con este nombre.
802-3-ethernet.mac-address . . .	HWADDR= . . .	La conexión se limita a la interfaz de red con esta dirección MAC.

/etc/resolv.conf

```
[root@rhel ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain lab.local
nameserver 10.200.0.1
[root@rhel ~]# nmcli connection up ESO
Conexión activada con éxito (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/8)
[root@rhel ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain lab.local
nameserver 10.200.0.1
nameserver 213.176.161.16
[root@rhel ~]# nmcli connection up CCFF
Conexión activada con éxito (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/9)
[root@rhel ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain casa.local lab.local
nameserver 10.200.0.1
nameserver 10.200.0.100
[root@rhel ~]# █
```



ESCENARIO DE TRABAJO

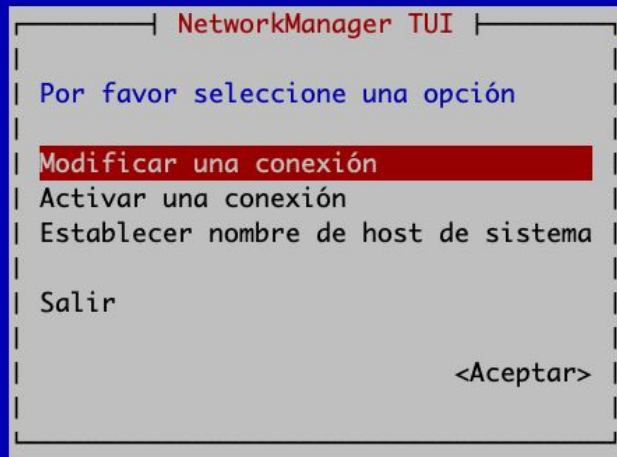
HERRAMIENTAS NECESARIAS

USO DE NMCLI

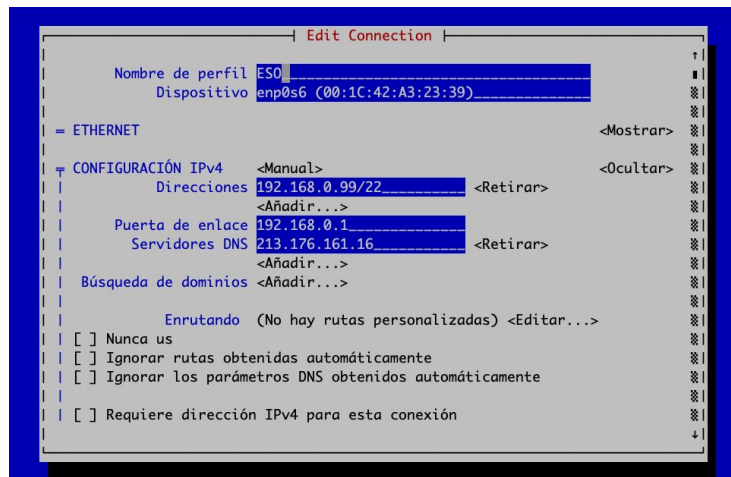
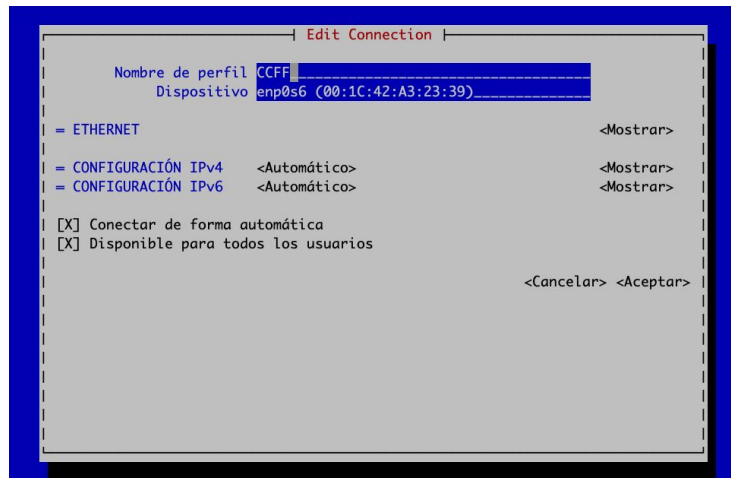
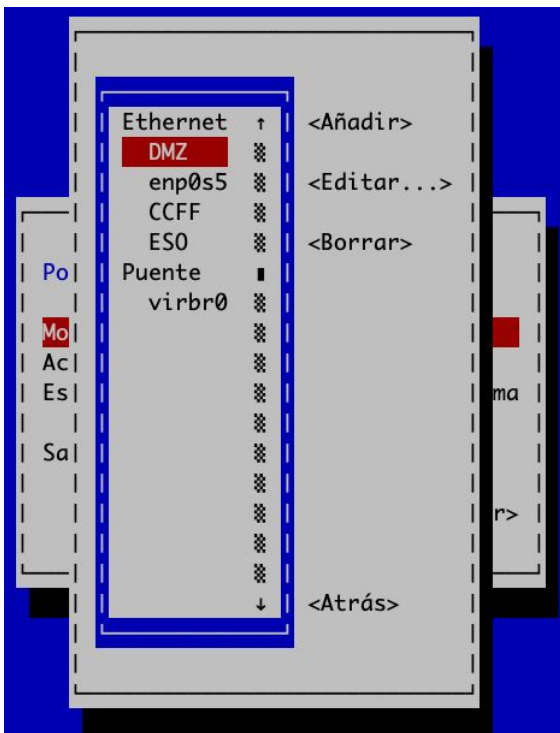
FICHEROS DEL SISTEMA

NMTUI

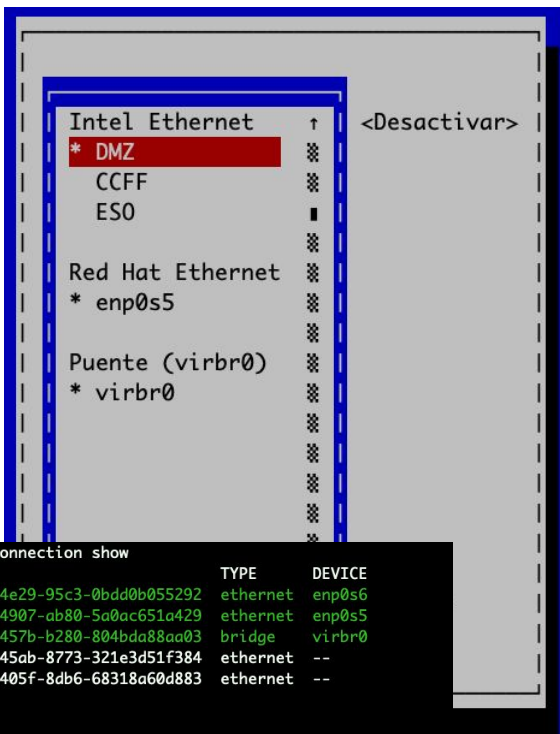
nmtui



nmtui - Modificar una conexión



nmtui - Activar una conexión

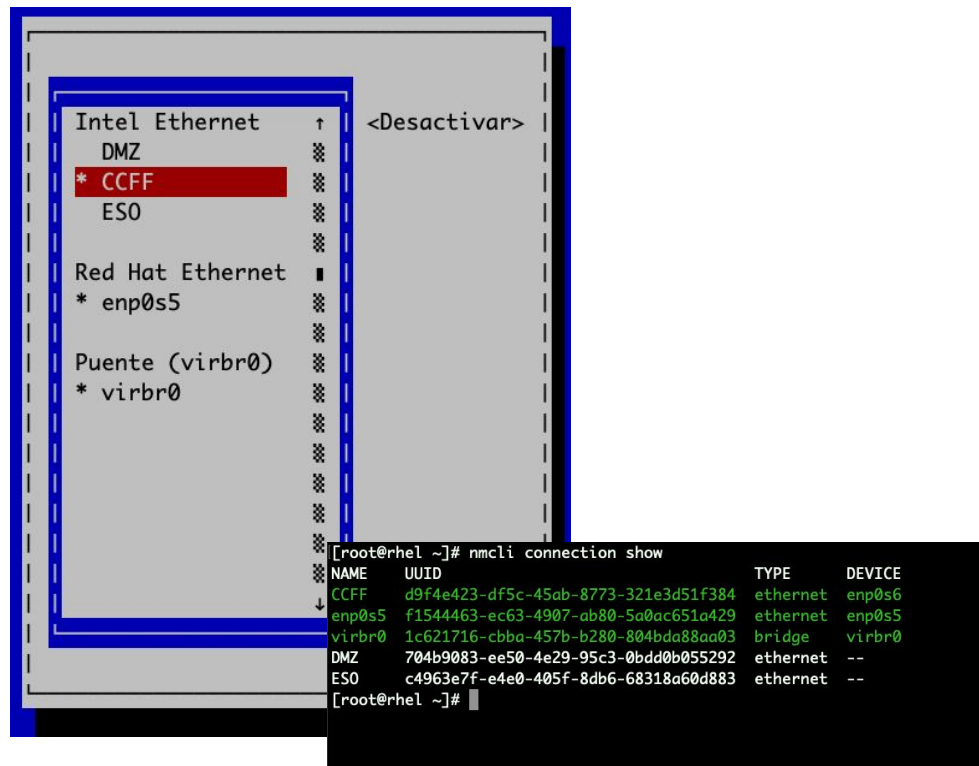


The nmtui interface shows a list of network connections. The 'DMZ' connection is highlighted in red, indicating it is selected. The interface is titled 'Intel Ethernet' and includes a '<Desactivar>' button. The connections listed are DMZ, CCFF, ESO, Red Hat Ethernet (* enp0s5), and Puente (virbr0) (* virbr0).

```
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
DMZ	704b9083-ee50-4e29-95c3-0bdd0b055292	ethernet	enp0s6
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
virbr0	1c621716-cbba-457b-b280-804bda88aa03	bridge	virbr0
CCFF	d9f4e423-df5c-45ab-8773-321e3d51f384	ethernet	--
ESO	c4963e7f-e4e0-405f-8db6-68318a60d883	ethernet	--

```
[root@rhel ~]#
```



The nmtui interface shows a list of network connections. The 'CCFF' connection is highlighted in red, indicating it is selected. The interface is titled 'Intel Ethernet' and includes a '<Desactivar>' button. The connections listed are DMZ, CCFF, ESO, Red Hat Ethernet (* enp0s5), and Puente (virbr0) (* virbr0).

```
[root@rhel ~]# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
CCFF	d9f4e423-df5c-45ab-8773-321e3d51f384	ethernet	enp0s6
enp0s5	f1544463-ec63-4907-ab80-5a0ac651a429	ethernet	enp0s5
virbr0	1c621716-cbba-457b-b280-804bda88aa03	bridge	virbr0
DMZ	704b9083-ee50-4e29-95c3-0bdd0b055292	ethernet	--
ESO	c4963e7f-e4e0-405f-8db6-68318a60d883	ethernet	--

```
[root@rhel ~]#
```



ESCENARIO DE TRABAJO

HERRAMIENTAS NECESARIAS

USO DE NMCLI

FICHEROS DEL SISTEMA

NMTUI

NMCLI - WIFI

NMCLI - Conexión a red Wi-Fi

```
[root@black:~] # nmcli connection show
NAME      UUID                                  TYPE      DEVICE
HOME      e3143a71-42c4-3152-b026-c29a8f4e4ac1 ethernet  enp0s20u1
CASA      0e006be5-9260-46b4-a614-102541f593d9 wifi      --
LAB       7e5f28d9-58f4-30db-bcc1-34927d492c8f ethernet  --

[root@black:~] # ip address show wlp2s0
4: wlp2s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether ee:4c:ae:3a:ce:98 brd ff:ff:ff:ff:ff:ff permaddr 18:cf:5e:d1:da:66

[root@black:~] # nmcli device wifi connect
0C:80:63:B9:DF:94      8E:24:CD:AC:B3:67      TP-Link_DF94          XAvi
78:81:02:94:21:71     A2:81:02:94:21:72     TP-LINK_Extender_3A248F xaviym

[root@black:~] # nmcli device wifi connect XAvi password Academy123
Device 'wlp2s0' successfully activated with '9051a422-f790-4ee6-936d-f865e0add548'.

[root@black:~] # ip -4 address show wlp2s0
4: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    inet 172.20.10.19/28 brd 172.20.10.31 scope global dynamic noprefixroute wlp2s0
        valid_lft 85399sec preferred_lft 85399sec

[root@black:~] # nmcli connection show
NAME      UUID                                  TYPE      DEVICE
HOME      e3143a71-42c4-3152-b026-c29a8f4e4ac1 ethernet  enp0s20u1
XAvi      9051a422-f790-4ee6-936d-f865e0add548 wifi      wlp2s0
CASA      0e006be5-9260-46b4-a614-102541f593d9 wifi      --
LAB       7e5f28d9-58f4-30db-bcc1-34927d492c8f ethernet  --
```



¡Muchas gracias!

 www.pue.es

 blog.pue.es/pue-academy

    [@pue_academy](https://www.instagram.com/pue_academy)

PUE Academy

 pueacademy@pue.es

Certification

 exams@pue.es

PUE Services

 sales@pue.es

PUE Training

 training@pue.es